

StatKarmayog Learning Material

Sampling Design and Data Quality Guidelines for Official Statistics

This document is a fictional training note prepared for testing AI-based quiz generation. It contains readable text about sampling, validation, data quality checks, and reporting practices used by statistical officers.

1. Purpose of Sampling Design

Sampling design defines how units are selected from a population so that survey results are reliable, unbiased, and cost-effective. A good design identifies the target population, sampling frame, stratification variables, sample size, and selection procedure. Statistical officers should document why each design choice was made.

2. Stratified Sampling

Stratified sampling divides the population into homogeneous groups called strata. Samples are then drawn from each stratum. This method improves precision when strata are internally similar and different from each other. Common strata in official statistics include district, sector, gender, rural-urban status, and establishment size.

3. Sampling Frame Quality

A sampling frame is the operational list from which sample units are selected. Frame quality affects coverage error. Officers must check duplicate entries, missing units, obsolete records, inconsistent identifiers, and incomplete address information before drawing the sample.

4. Sample Allocation

Sample allocation determines how many units are selected from each stratum. Proportional allocation assigns samples according to stratum size. Optimum allocation may consider both stratum size and variability. The chosen allocation method should match the survey objective and available resources.

5. Data Validation Rules

Data validation rules detect errors before analysis. Range checks identify values outside permitted limits. Consistency checks compare related fields. Duplicate-key checks detect repeated unique identifiers. Missing-value checks flag unanswered mandatory questions. These checks should be automated wherever possible.

Example Validation Matrix

Validation Check	Purpose	Example
Range check	Detect impossible values	Age must be between 0 and 120
Duplicate-key check	Find repeated records	Household ID appears twice
Consistency check	Compare related answers	Employment status and hours worked
Missing-value check	Identify blank mandatory fields	District code is empty

6. Data Quality Dimensions

Data quality is assessed through accuracy, completeness, consistency, timeliness, validity, and uniqueness. Accuracy checks whether data represents the real-world condition. Completeness measures whether required data is present. Consistency ensures related fields do not contradict each other.

7. Treatment of Missing Values

Missing values should be classified before treatment. Unit non-response occurs when an entire sampled unit provides no data. Item non-response occurs when only selected fields are missing. Treatment options include follow-up, imputation, flagging, or exclusion, depending on the analysis requirement.

8. Outlier Review

Outliers are observations that differ substantially from the rest of the data. They may represent true rare events or data errors. Statistical officers should review outliers using domain knowledge, supporting documents, and comparison with historical values before changing or excluding records.

9. Documentation and Metadata

Every survey dataset should include metadata describing definitions, units, source, collection period, validation rules, revision history, and limitations. Good metadata makes statistical outputs easier to interpret, audit, and reuse.

10. Reporting Findings

A statistical report should explain methodology, data limitations, key findings, charts, interpretation, and quality caveats. Officers should avoid unsupported conclusions and clearly distinguish observed evidence from policy interpretation.

Assessment Focus Areas

After reading this document, a learner should be able to explain stratified sampling, identify sampling-frame errors, select appropriate validation checks, distinguish types of missing values, review outliers responsibly, and describe why metadata is essential for official statistics.

Suggested Quiz Competencies: Sampling, Survey Design, Data Quality Frameworks, Metadata Standards, Data Visualization, and Communication.